

DES Pune University, Pune
School of Engineering and Technology
Department of Computer Engineering and Technology
Program: B. Tech in Computer Science and Engineering

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Subject: Python Programming

Assignment No.: 16

Date: 02/02/2026

Lab Assignment: 16

Aim:

Write a program to calculate GCD using recursive functions.

Objective:

To understand and implement recursion using Euclid's algorithm.

Theory:

GCD (Greatest Common Divisor) is the largest number that divides two numbers exactly.

The GCD is calculated recursively until the remainder becomes zero.

Algorithm:

- Start
- Define a recursive function for GCD
- If second number is 0, return the first number
- Otherwise, call the function again
- Display the GCD
- Stop

Pseudo Code:

```
FUNCTION gcd(a, b)
IF b == 0
    RETURN a
ELSE
    RETURN gcd(b, a % b)
END IF
END FUNCTION
```

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Code:

```
def gcd(a, b):  
    if b == 0:  
        return a;  
    else:  
        return gcd(b, a % b);  
num1 = int(input("Enter first number: "));  
num2 = int(input("Enter second number: "));  
result = gcd(num1, num2);  
print("GCD of the two numbers is:", result);
```

Output:

```
Enter first number: 21  
Enter second number: 42  
GCD of the two numbers is: 21
```

Conclusion:

The program successfully calculates the GCD of two numbers using recursion.